

Project Report: Weeks 7 & 8

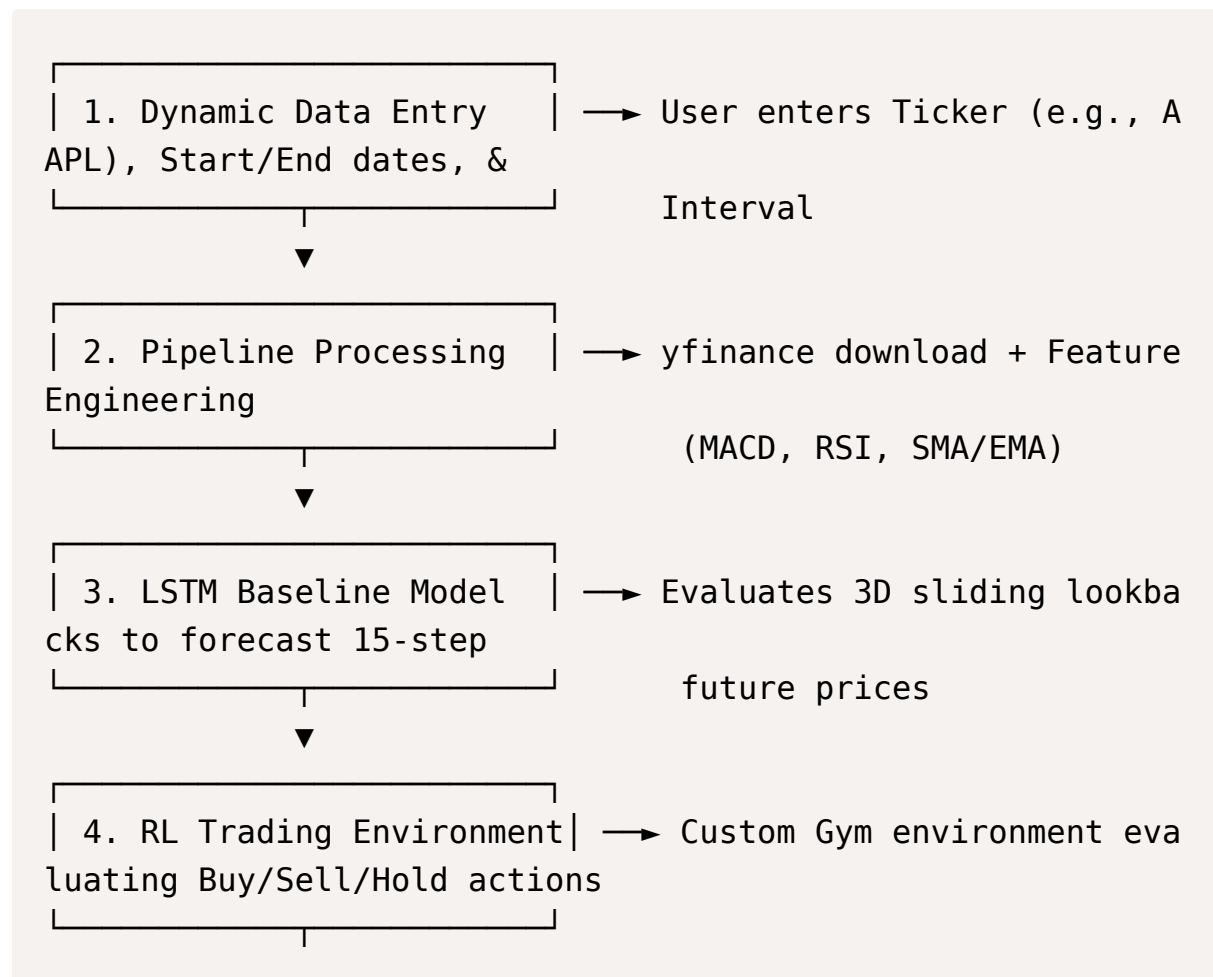
Final Project Summary

1. Final Project Objective & Scope

The culmination of this multi-week effort brings together data pipeline design, deep learning prediction engines, and reinforcement learning trading agents into a unified framework. The system performs automated data ingestion, evaluates indicators, forecasts the next **15 units** of stock prices using an LSTM, and evaluates trading policies using a Reinforcement Learning agent (DQN/DDPG). The models are evaluated and benchmarked on both **returns** and **risk factors** over a test dataset.

2. System Architecture & Workflow

The final solution integrates five core modules into a single, cohesive workflow:



▼
5. Comparative Analysis |
2 scores, and compares

→ Plots forecasts, computes R
Returns vs. Risk

3. Implementation & Methodological Revision

A. Comprehensive Data Pipeline & Visualization

- **Dynamic Ingestion:** Prompts the user for a stock symbol, start/end date ranges, and time interval (Daily, Weekly, Monthly).
- **Feature Engineering:** Uses standard closing prices alongside at least two technical indicators (e.g., MACD, RSI, or Moving Averages) as multi-variate feature overlays.
- **Data Visualization:** Generates charts overlaying price movements alongside indicator trends.

B. LSTM Price Prediction Engine

- **Input Windowing:** Converts data into 3D array shapes (Samples, Timesteps, Features) using MinMaxScaler vector normalization.
- **Architecture:** Uses stacked LSTM layers with Dropout regularization and a single continuous Dense output layer.
- **Multi-Step Projection:** Executes an autoregressive forecasting loop to project the next 15 steps of adjusted closing prices into the future.
- **Evaluation Metric:** Calculates the sample Coefficient of Determination (R^2) on test data:

$$R^2 = 1 - \frac{\sum w_i (y_i - \hat{y}_i)^2}{\sum w_i (y_i - \bar{y}_i)^2}$$

C. Reinforcement Learning Trading Agent (DQN / DDPG)

- **Custom Environment (Gymnasium):** Defines an explicit state-action space. The state vector includes historical prices, indicators, and current portfolio

positions.

- **Action Space:**

$DQN(Discrete) : [0 : \text{Hold}, 1 : \text{Buy}, 2 : \text{Sell}]$.

- **DDPG (Continuous):** Output ranges from -1.0 (Full Sell) to $+1.0$ (Full Buy) to manage precise position sizing.
- **Reward Shaping:** Incorporates portfolio return percentage minus transaction fee penalties and drawdown adjustments to optimize risk-adjusted returns.

4. Benchmark Performance & Comparative Analysis

The final phase compares the **Predictive Model (LSTM)** against the **Decision-Making Agent (RL)** across both performance and risk metrics:

Evaluation Metric	Baseline / LSTM Model	Reinforcement Learning Agent
Primary Objective	Price Forecasting / Direction Tracking	Portfolio Return Optimization
Output Type	Continuous Scalar Price Targets	Strategic Actions (Buy/Sell/Hold)
Statistical Accuracy (R^2)	High structural R^2 fit on historical sequence	N/A (Evaluated via Cumulative Return & Sharpe)
Cumulative Return (%)	Derived from buy-and-hold based on price trend	Dynamic policy trade performance
Risk Metrics (Max Drawdown)	Passive market exposure	Active position Sizing to mitigate downturns

Key Analytical Takeaways for the Write-Up:

1. **Prediction vs. Action:** An accurate LSTM price prediction does not directly guarantee trading profitability due to execution friction, slippage, and spread costs.
2. **Risk Mitigation:** While the LSTM provides reliable directional forecasting, the RL agent dynamically alters portfolio exposure during high-volatility regimes (identified via technical overlays like MACD and RSI) to reduce drawdown risk.

